12

The Development of Deductive Reasoning

Henry Markovits

The ability to make deductive inferences is one of the defining characteristics of advanced human thinking. Being able to take some given premises as a base and then to explore the kinds of necessary deductions that can be made from them is a critical component of mathematics and science. More broadly, making inferences underlies much of our day-to-day thinking in varied contexts (e.g., Lea, O'Brien, Fisch, Noveck, & Braine, 1990). Understanding how this ability develops is a correspondingly important question, both for theoretical and practical reasons. However, the study of the development of deductive reasoning presents some major problems. As we shall see, there is not much consensus about just what children and adults can actually do, or about what, if anything, develops in deductive reasoning.

Underlying at least some of the problems that are present in this domain is an important question concerning the status of logic in the study of reasoning. There is a clear distinction to be made between deductive reasoning and logical reasoning (Overton, 1990). Deduction refers to the process of reaching a conclusion on the basis of some given premises. Logic is a formal domain that attempts to characterize specific forms of argument as being valid or invalid. In other words, logic provides textbook norms for correct deductions, norms that are often explicitly used in academic contexts to evaluate the adequacy of deductive reasoning. The importance of logical norms to describe arguments is so intuitively clear that the expression "not being logical" is a commonly used way of describing a person whose arguments are considered to be bad. Whether logic is considered to

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be a model for the human mind, as did Boole (1854), one of the founders of modern logic, or whether it is simply used to provide cues as to what constitutes a correct response to a deductive problem, the presence of logic as a normative system has had an important effect on both theoretical and empirical explorations of reasoning.

This has specifically affected developmental studies of reasoning in several ways. The most important of these is the fact that many, if not all, developmental studies have taken logical reasoning as the baseline for evaluating reasoning performance. The results of most developmental studies can be synthesized by one or the other of the following two descriptions: Children are logical or adolescents are more logical than children. Logical reasoning is thus, at least implicitly, seen as either the starting point or the end point of development. It is telling in this respect to compare developmental studies to those that look specifically at reasoning in adults. The results of a large majority of studies of adult reasoning can be synthesized by the description: "Adults are not logical." This has, in turn, prompted many researchers to raise the question of whether or not people are indeed able to make normative inferences, or in other words, whether people are, or even can be, rational (Manktelow & Over, 1993). It is clear that developmental researchers and those examining reasoning in adults have generally come to very different conclusions about the relationship between logic and reasoning. I will not make any specific suggestions at this point, but it is worth remarking that developmental and adult studies do not examine different species, and that interpretations of the outcomes of these studies must, in the end, be mutually compatible.

As I have previously suggested, data on how children and adults perform on deductive reasoning problems have been interpreted in very different ways. On the one hand, empirical work has shown that educated adults have great difficulty in making logical inferences on some basic forms of reasoning (Evans, 1982). Conversely, other studies appear to show that very young children can make logical inferences on these same basic forms of reasoning (Ennis, 1976; Hawkins, Pea, Glick, & Scribner, 1984). As Moshman (1990) has so aptly remarked, this pattern of results suggests that reasoning competence peaks in early childhood and rapidly declines afterward (which anyone who has tried to raise young children may not find surprising). One of the challenges of any developmental theory is to explain this apparent contradiction. This chapter will start with an overview of some of the major theories that have attempted to explain the nature and development of logical reasoning (some of these theories will necessarily overlap with content from other chapters). Subsequently, certain key results of empirical studies that have looked at reasoning in a developmental context will be examined. Finally, I will present a recent approach based on mental-model theory (Johnson-Laird, 1983) that attempts to account for

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both early competence and later incompetence within a specific procedural model.

LOGICAL REASONING IN PIAGET'S THEORY

Inhelder and Piaget (1958) were among the first researchers to systematically examine the development of deductive reasoning. The capacity to make logical deductions was seen as one of the defining characteristics of formal thinking, which was considered to be the final stage of cognitive development. As such, a formal operational thinker was considered to have the basic competence to reason in a way that was consistent with basic propositional logic. This did not imply that inferential reasoning was beyond the ability of younger, more concrete thinkers. In fact, Piaget (Piaget, Grize, Szeminska, & Bang, 1977) considered that making inferences is a fundamental part of cognition at all levels. For example, a child who claims that if liquid in a container is emptied into another container of different form, then the height of liquid in the two containers will be identical is making an inference. What distinguishes these, more primitive forms of inference from formal deductive inferences is a major part of Inhelder and Piaget's qualitative description of formal thinking. Since this description has rarely been exploited in subsequent research, it is worth giving it some degree of attention.

One of the key notions used to characterize formal thinking is the idea that formal inferences are not made in isolation, but can be combined in an inferential structure, where each inference has an impact on other components. Thus, a formal reasoner has the capacity to perform the kind of complex reasoning that would allow a given conclusion to impact on previously accumulated information (e.g., I can conclude that P must be true from these premises, but observation X suggests that P is false, etc.). In contrast, more concrete modes of reasoning act in a sequential manner, such that the conclusion of one inference remains unrelated to the conclusion made from a previous inference. Younger reasoners can be easily lead to making what appear be to internally contradictory conclusions (see also Piaget, 1974). A second important characteristic of formal thought (which is related to the previous point) is the nature of the relationship between a deductive inference and empirical observation. Concrete reasoning is essentially led by observation, both because observation is important in order to specify the nature of the concepts that are used in reasoning and, more importantly, because any conflict between inference and observation will tend strongly to be resolved in favor of observation. Formal reasoning is characterized by an inversion of this relationship. When a formal reasoner is faced with a conflict between a deductive inference and observation, he or she will tend to put relatively more weight on inference than on observation. This is because a formal inference can be seen as being

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part of a larger structure that considers multiple interrelations between observation and conclusions. Thus, when a formal inference is (correctly) made, it must not only be locally consistent with the immediate premises, but also globally consistent with previous conclusions and observations. For example, when a logical conclusion is reached that "P must be true," this conclusion has implications for a variety of other components that include other observations. Thus, if an observation is made that suggests that "P is false," this observation might put into doubt, not only the immediate conclusion, but also other empirical observations. In this case, the formal reasoner might well be prepared to question the observation more readily than the logical conclusion. A final point is the relation between necessity and what is referred to as possibility. A logical conclusion is generally defined as a necessary one. Simply put, P is a necessary conclusion from A if, given premises A, it is not possible that P is false. What Inhelder and Piaget (1958) claimed was that the underlying necessity of logical inferences was a function of the ability of a reasoner to conceive of a more or less greater range of possibilities (see also Piaget, 1987a, 1987b). The degree of necessity of a given conclusion depends on how many of the potential alternatives are accounted for in the course of making a given inference; a conclusion is truly necessary when it takes into account all possible alternatives. Thus, they considered that one of the chief underlying abilities for formal thought was the ability to systematically (and combinatorially) generate all possible states in any given situation. Since younger, more concrete, thinkers are generally not able to generate all possibilities, their inferences show what might be called local necessity, since the same inference might be judged to be not necessary if different possibilities are considered by the same reasoner.

Inhelder and Piaget (1958) also attempted to provide a characterization of what a formal reasoner was ideally able to do, that is, a description of the basic competencies underlying formal thinking. Generally, they claimed that standard propositional logic was an appropriate competence model for formal reasoners. Specifically, they stated that a formal reasoner should be able to correctly use the 16 binary operations composed of all possible combinations of (p.q, p.not-q, not-p.q, not-p.not-q). For example, the conditional *if p then q* can be defined by a truth table for which p.q, not-p.q, not-p.not-q are true and p.not-q is false. The most straightforward interpretation of this claim is that formal reasoners have inferential rules that correspond to all of these combinations (e.g., rules for making conditional inferences). (It should be noted that a fair interpretation could less stringently claim that formal reasoners have unspecified cognitive processes that allow them to make systematically correct inferences to these logical forms in ideal conditions). Concrete reasoners, in contrast, can occasionally make some of these inferences correctly, but not systematically, and only if aided by some kind of concrete support.

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Now, the most direct claim that can be made on the basis of this analysis is that only adolescents and adults should be able to reliably make correct inferences to verbally presented inferential problems. Another related claim is that most adolescents and adults should be able to do this, although this latter is less compelling. Following Inhelder and Piaget's (1958) analysis of formal thinking, there was a concerted effort to examine these claims, mostly using logical inferences related to conditional (if-then) reasoning. Early results were somewhat inconsistent with these two predictions. For example, studies did show that before adolescence, the proportion of reasoners who could give correct responses to the four logical forms that characterize conditional reasoning was very small (Wildman & Fletcher, 1977). However, children as young as six or seven years of age could make correct inferences to verbal problems on certain of the logical forms (Ennis, 1976). In addition, many studies showed that even adults had great difficulties in making logically correct inferences (e.g., Markovits, 1985). One of the more striking of these was a seminal study by Wason (1968), who presented a group of highly educated students with a task based on conditional logic, referred to as the selection task. The proportion of these reasoners who gave the correct response was remarkably low and led Wason to conclude that Piaget's analysis of reasoning was incorrect. Whatever the interpretation of these results is, it is clear that there is a great deal of variability in inferential performance at all ages. It is this fact that underlies Overton's competence-performance model.

OVERTON'S COMPETENCE-PERFORMANCE MODEL

The first studies that examined the more strict interpretations of Piaget's model suggested that these could not really be sustained in their initial form. Overton (Overton, 1985, 1990; Overton & Newman, 1982) subsequently presented a model of reasoning that relies on the distinction between competence and performance factors in order to explain this initial pattern of results. Specifically, this model claims that Piaget's basic analysis was correct, that is that younger children do not have the basic competence to make logically correct inferences, while adolescents and adults do. However, the inconsistency in performance that is often observed with older reasoners can be accounted for by performance factors of various kinds that do not allow the full expression of this competence. Simply put, this model claims that adolescents and adults generally possess cognitive processes that, when properly deployed, allow them to make consistently logical inferences to deductive reasoning problems. However, various factors that are unrelated to these basic processes result in their being used in a suboptimal way, or not being used at all. Among such factors are classical ones such as fatigue and memory load and others that are more fundamental, such as familiarity (see also Piaget,

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1972, for a more general analysis of how familiarity might affect formal reasoning).

This model allows for two related predictions. The first relies on the claim that poor inferential performance in children and in older reasoners does not have the same source. Younger reasoners do not possess the basic competence (i.e., the requisite cognitive processes) required to make logically correct inferences. Older reasoners do possess these cognitive processes, but may not show their competence because of various performance factors. What would then happen if younger and older reasoners who are equally hapless in initial inferential performance were helped out in some way that should cue improved performance. Helping younger reasoners should be relatively futile, since they do not have the requisite competence to profit from any intervention. However, this same intervention should be quite profitable in the case of older reasoners. This particular prediction was indeed confirmed in a set of studies (O'Brien & Overton, 1980, 1982; Overton, Byrnes & O'Brien, 1985) that used a procedure by which reasoners were given information designed to contradict a faulty interpretation of a conditional rule. Generally, this intervention was effective in improving performance among reasoners at grade 12 and older, but ineffective before this age. Another, related, prediction that Overton's model allows concerns the effects of variables such as familiarity and the related concept of relevance (Ward & Overton, 1990). Generally, within the competence-performance model, familiar and/or relevant material is considered to facilitate reasoning, since it provides cues that activate appropriate cognitive procedures, when these are available. In the case of younger reasoners who do not possess this basic competence level, familiarity would not do this, since these procedures are not available. Once again, in a series of studies that examined performance on a modified version of the selection task, this prediction was confirmed (Mueller, Overton, & Reene, 2001; Overton, Ward, Black, Noveck, & O'Brien, 1987; Ward & Overton, 1990). In this case, familiarity resulted in improved performance only among older reasoners, although at an earlier age than was observed with the previous studies.

This model provides a useful counterpoint to the Piagetian model that serves as its base. It makes quite explicit the idea that inferential performance is highly variable, even among educated adults who are, at least theoretically, at a high level of formal competence. It presents the important idea that this variability can be understood in an interactive way as a function of a reasoner's initial level of competence.

Overton's model specifically addresses the question of the status of *logically incorrect* responses produced by reasoners who are expected to be at a high level of competence. In addition, there is another class of problematic responses that also raises questions for the Piagetian model. These are cases in which quite young reasoners can consistently give the logically correct response to at least some deductive reasoning problems (Ennis,

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1976; Hawkins et al., 1984; Kuhn, 1977). Results of this kind have been interpreted in differing ways. Piagetian theorists claim (Byrnes & Overton, 1986; Knifong, 1974; Overton, 1990) that a major criterion for formal reasoning involves the general structure of the inferences that are made. This position supposes that processes that are not truly formal could lead to logically correct responding to one or two isolated logical forms. For example, it has been claimed that young children can use various surface-level strategies that lead to logically correct responses to simple inferences (Knifong, 1974; Markovits, Schleifer, & Fortier, 1989). It is the ability to respond logically to a wide range of inferential problems that truly corresponds to formal reasoning.

However, there are alternative approaches to reasoning that consider that the capacity to give the logically correct answer to a limited subset of inferential problems should, in fact, be considered as an indication of some basic competence. The best known of these is Braine's theory of natural logic (Braine, 1978; Braine, 1990; Braine & O'Brien, 1991; Braine & O'Brien, 1998).

BRAINE'S THEORY OF NATURAL LOGIC

Braine's theory of natural logic makes two major claims. The first part of Braine's theory is the claim that reasoning involves the use of specific inferential schemas. In this respect, this theory resembles Piagetian theory (at least on a strict interpretation of the latter). However, Braine did not, as Piaget did, use standard propositional logic as a model for human reasoning. Natural or mental logic is seen as incorporating a subset of basic rules that are used to make real-life inferences. While most of these correspond to standard logical norms, a few do not. Thus, the mental logic model, while in most ways consistent with standard logic, does allow for some incorporation of reasoners' unschooled intuitions.

Braine's analysis of the developmental trajectory of reasoning competence is quite different from the Piagetian analysis. There are some important reasons for this difference. In Braine's theory, there are no interconnections between different inference schemas. More importantly, Braine (1990) makes the strong claim that some of the inferential rules that appear to be available to young children (which he refers to as primary reasoning skills) are biologically based. The specific argument is that certain inferences are so critical to behavior, particularly with respect to language comprehension and the integration of information derived from diverse sources, that the selection pressure to incorporate them into the basic architecture of the human mind would have been very strong (see also Cohen, 1981; MacNamara, 1986). These inferential procedures involve direct inferences that are controlled by a simple program. One important example is the modus ponens inference, which is considered to be one of

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the key rules, since it allows the kind of elementary inference that would be critical to basic communication (e.g., "If a lion, then danger. There's a lion. There must be danger"). Other primary schemas include those for negation, and basic connectives such as *and* and *or* (see Braine & O'Brien, 1998, p. 80–81 for a complete list). More complex inference schemas are referred to as indirect reasoning strategies or secondary skills. These generally involve some form of recursive reasoning. The classic *reductio ad absurdum* argument (this is an argument form that requires assuming the falseness of the proposed conclusion in addition to the premises and then proving that this combination leads to a false conclusion) is an example of one of these schemas. Indirect schemas are acquired with training and experience, particularly experience with language and in academic contexts.

This theory thus supposes that some of the basic inferential rules required for making deductive inferences are a part of a biologically determined cognitive architecture. Inferential schemas of the kind that are postulated here have a form that relies on the basic syntax of the particular deduction that is required. Thus, the rule for modus ponens looks like:

$$\begin{array}{l} \text{if } p \text{ then } q;\ p \\ \quad\ q \end{array}$$

Theoretically, the conclusion to any inference that shares this basic structure will be formally identical, since all such inferences will be mapped onto the same rule. Does this mean that these basic inferences will always be correctly made? Not necessarily. The natural logic model explicitly recognizes the fact that untrained reasoners usually receive inferential problems that are phrased in ordinary language. The model contains an initial, interpretative stage that allows for a semantic analysis of premises. This initial evaluation invokes various pragmatic principles that can in fact alter the way that a real-life inferential problem is actually solved. There are, at least, two levels to this evaluation. The first level concerns the global form of processing that is to be used by the reasoner. Following previous work by Luria (1971) and Scribner (1977), Braine (1990) distinguishes between natural reasoning and analytic reasoning. Natural reasoning is reasoning that invokes the full range of the reasoner's knowledge about whatever phenomena are referred to in a given inference, in order to derive and evaluate a conclusion. Analytic reasoning is reasoning that only uses the specific syntactic information given in an inferential problem to arrive at a conclusion. Natural reasoning is thus reasoning for which the logical status of a possible conclusion is not the only important element in evaluating the conclusion. For example, if a logical conclusion violates strongly held beliefs, then these beliefs may well (and reasonably) take precedence over logical validity. In contrast, analytic reasoning looks only at the syntax of a given problem and suppresses use of beliefs. Thus, one level of

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interpretation that is involved in analyzing an inferential problem is determining the overall goal of the problem, specifically, whether the reasoner should take an analytic or a natural stance when making a conclusion

There is a second level of interpretation that is also invoked. This concerns the possible use of pragmatic or conversational principles (Grice, 1975) that may modify the interpretation of a given logical form. One classical example is the use of if-then in the context of promises and threats. Statements such as "If you eat your supper, you'll have dessert" are commonly used in ordinary conversation. Although such statements contain an if-then surface appearance, they are in fact usually intended to be effective biconditionals (if and only if) (Fillenbaum, 1975, 1976). In the previous example, the intention of the speaker is clearly to imply that "If you don't eat your supper, you won't have dessert." A strict conditional interpretation of this utterance would allow for the possibility of not eating supper and still having dessert, something that would certainly reduce the effectiveness of the underlying threat implied by the initial statement. Thus, it is clear that in many circumstances, reasoners make implicit inferences on the basis of pragmatic principles that may alter the underlying logical structure of a given statement. Thus, one goal of any interpretative module is to establish which specific inferential rule is intended in a given utterance. One key developmental trend that has been analyzed in this way is the strong tendency of younger children to respond to conditional statements as if these were biconditionals (Wildman & Fletcher, 1977). Braine's model suggests that any tendency to misinterpret conditional statements should be due to children's generating inappropriate implicit inferences. Rumain, Connell, and Braine (1983) hypothesized that if young children were given supplementary cues that contradicted any implicit biconditional inference, then they should be able to respond correctly to the two uncertain logical forms of the conditional. They did indeed find this to be the case, and accordingly claimed that the general developmental improvement in performance on these forms is due to the increased facility with which older reasoners can set aside such implicit inferences. Interestingly, a differing developmental prediction based on the same notion of pragmatic inferences was made by Cahan and Artman (1997). Their basic argument was that since the effects of pragmatic inferences depend mainly on experience, there should actually be a developmental increase in the tendency of reasoners to respond biconditionally. They used an ingenious technique to attempt to dissociate large-scale developmental trends (which could be due either to learning and/or to developmental processes) from the more local effects of increased experience with pragmatic inferences, and did indeed find evidence for a developmental tendency to increased biconditional responding.

Overall, Braine's theory considers that evidence of early competence reflects the precocious existence of certain basic deductive schemas that

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are, on the whole, logical. Development follows basically through training and experience and consists of subsequent acquisition of more complex schemas and the increasing ability to play the logical game. One key component of development is the acquisition of an analytic stance, that is the ability to reason strictly on the basis of given premises without using available real-world knowledge (although see Leevers & Harris, 1999 for an alternate view on this issue). Another component would involve learning of the more complex indirect schemas. This could be tied to specific training, or, as suggested by Falmagne (1980, 1990), to experience with language. A final component would be the ability to increasingly discount various pragmatic influences on reasoning, which would allow reasoners to behave increasingly logically, at least when the appropriate inference schemas were part of their cognitive repertoire.

METACOGNITIVE DEVELOPMENT

Another response to the question of what characterizes development, if young children are indeed assumed to have a basic ability to make logically appropriate inferences, has been given by Moshman (1990, 1995, 1998). His analysis of the empirical results that have been briefly cited makes two points. First, he accepts that young children do have the competence to make simple logical inferences. He also notes that much of the adult literature requires reasoners to make more complex inferences under sometimes difficult circumstances. Moshman makes a distinction between logic, which is seen as the capacity to make simple inferences, and metalogic, which involves a more explicit metacognitive awareness of a reasoner's own logical reasoning processes. He accepts Braine's argument that some basic inferential schemas are available to very young children, and then claims that further development is characterized by the increasing ability to reflect upon one's own reasoning, that is, that the major advances in reasoning ability beyond early childhood are mediated by greater metalogical abilities. Moshman (1990) also makes the strong claim that metalogical development progresses toward greater rationality. He distinguishes four major stages of metalogical development. The first stage is characterized by the ability to make implicit inferences. The second is characterized by the use of explicit inferential procedures and by an implicit understanding of logical distinctions. This kind of reasoning appears at about age six and develops throughout elementary school. The third stage consists of the ability to make explicit logical distinctions, something that appears at around 11 years of age. Explicit understanding of a variety of concepts, including the distinction between logical validity and empirical truth (Moshman & Franks, 1986), and the nature of tautologies (Osherson & Markman, 1975), are not explicitly understood before 11 years of age (but see Morris, 2000), but appear to develop in older reasoners. The

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fourth stage represents the basic culmination of metalogical development (although further analyses have led to the idea that there might be more than four stages) and is characterized by the ability to reflect upon logic as a formal system.

WHAT DO EMPIRICAL STUDIES SHOW ABOUT THE DEVELOPMENT OF REASONING ABILITIES?

Despite their common evocation of logic as a competence model for reasoning, these theories make quite divergent predictions about the way that children and adults can be expected to reason. To what extent do empirical studies that have examined deductive reasoning in children and adults support these various conceptions of development? Each of the presented theories uses a selected subset of the many studies that have examined reasoning in order to support its particular predictions, while often either ignoring other studies or claiming that their results are inconclusive for various reasons. Given this, it seems particularly useful to present a resume of some of the principle findings of studies that have looked at performance on two key deductive reasoning problems, with a particular emphasis on the contradictory interpretations that these studies might allow.

Is There Evidence that Young Children Can Reason "Logically"?

One of the key claims of Braine's theory is that there are some inferential rules that are so basic that they will, at the very least, develop quite early. The most studied of these is probably the modus ponens inference. This involves starting from the premises "if p then q, p is true" and concluding that "q is true." Some of the cited studies use syllogisms of the form "all X are Y, A is X, thus A is Y." Since both propositional and syllogistic forms appear to lead to quite similar response patterns (Markovits, Venet, Janveau-Brennan, Malfait, Pion, & Vadeboncoeur, 1996; Roberge & Paulus, 1971) no distinction will be made between them. Now, several studies have indeed shown that children as young as five years old correctly make the modus ponens inference either when familiar content is used in the premises (Ennis, 1976) or when abstract content is used (Hawkins et al., 1984). Young children do have problems making this inference when contrary-to-fact premises are used (Dias & Harris, 1988, 1990; Hawkins et al., 1984; Markovits & Vachon, 1989). For example, if given premises such as "if it is raining, then the grass is dry," young children show a strong tendency to conclude that "it is raining" implies that "the grass is wet." However, when these children are given some external support, either by embedding the premises into a fantasy context (Dias & Harris, 1988, 1990; Markovits & Vachon, 1989) or by giving instructions to focus on the premises (Leevers & Harris, 1999), they consistently make the logical

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modus ponens inference, even with contrary-to-fact premises. These results show that when given appropriate instructions to "suppose that the premises are true" and some, fairly simple additional support, children as young as five years old can make the modus ponens inference quite consistently with a variety of familiar, abstract or contrary-to-fact premises.

The above results are certainly consistent with the idea that quite young children do have the logical competence required to make the modus ponens inference. Now, if very young children possess the competence to reason logically on modus ponens, then one would certainly expect that older children and adults would become increasingly expert in this form of reasoning, and that there should be a generalized increase in logically correct responding to the modus ponens inference. Evidence that this was not the case would be disconcerting, at the least. To put this into a different perspective, the hypothesis that young children can acquire the rule that $1 + 1 = 2$ and use it in many circumstances would reasonably lead to the assumption that this judgment would be made with increasing frequency in older children. The (probably false!) observation that many adults think that $1 + 1 = 3$ would certainly make the description of what young children are really doing when they respond that $1 + 1 = 2$ more problematic.

There are, in fact, a number of studies that suggest that modus ponens reasoning does not show the kind of generalized consistency that one would expect from the existence of early competence. Some studies have shown a developmental decrease in logically correct responding to modus ponens (Janveau-Brennan & Markovits, 1999; Markovits, 2002; Markovits, Fleury, Quinn, & Venet, 1998) when reasoning with causal conditionals ("if cause p then effect q"). Adolescents will deny the modus ponens inference when the logical conclusion is unbelievable (Markovits & Bouffard-Bouchard, 1992). Many studies have also shown that even educated adults have problems making this inference when presented with premises that are not believable (George, 1995, 1997), when responding to premises for which there are many possible ways that the antecedent might be true with the consequent being false (Cummins, 1995; Cummins, Lubart, Alksnin, & Rist, 1991; Thompson, 1994), when given additional premises with the same consequent (Byrne, 1989; Chan & Chua, 1994) or when they are simply asked to produce an alternative antecedent (Markovits & Potvin, 2001).

The studies that have examined reasoning on modus ponens thus present a varied profile. Some of the results are consistent with the conclusion that "young children are logical" and others, with the conclusion that "adolescents and adults are not logical." At this point, I will simply note that the available empirical evidence is globally inconsistent, and that reconciling the large variation in performance observed in adolescents and adults with the hypothesis that young children possess some basic logical competence appears difficult at best.

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Is There Evidence that Only Older Adolescents Can Reason "Logically"?

The Piagetian or neo-Piagetian approach would in contrast, claim that the basic competence required for formal reasoning is the result of a developmental process and only appears during later adolescence. Studies starting from this perspective have typically examined more complex forms of reasoning than that involved in modus ponens (MP). The most frequently studied forms are the two invalid inferences of conditional logic (which are often grouped together since they present a similar developmental pattern). The first of these inferences is affirmation of the consequent (AC). This corresponds to the following premises: "if p then q, q is true." The second such inference is denial of the antecedent (DA). This corresponds to the following premises: "if p then q, p is false." Neither of these logical forms leads to a single, necessary conclusion. The most frequent logically incorrect response in both cases is to make a biconditional inference, that is, to respond to these forms as if they used if-and-only-if as a connective. Specifically, for the AC form, the most frequent nonlogical response is to conclude, "p is true." For the DA form, the most frequent nonlogical response is to conclude, "q is false." Several studies have in fact shown clear developmental trends in reasoning with the two invalid forms, affirmation of the consequent and denial of the antecedent. Many studies have shown that children younger than 12 years of age produce a very high proportion of logically incorrect conclusions to both of these forms, and that there is a general increase in the proportion of logically correct responses with age (Klaczynski & Narasimham, 1998; Markovits & Vachon, 1990; O'Brien & Overton, 1980, 1982; Taplin, Staudenmayer, & Taddonio, 1974; Wildman & Fletcher, 1977). These results certainly appear to be consistent with the idea that logical competence is not accessible before adolescence and that it is the product of a developmental process.

However, there are two sets of results that raise some problems with this initial interpretation. On the one hand, some studies have shown that children as young as six or seven years old, can, in some cases, reliably make logical inferences to both affirmation of the consequent and denial of the antecedent with purely verbal premises (Markovits, 2000; Markovits et al., 1998; Markovits et al., 1996). Other studies have shown that children as young as seven years of age can give the logically correct response to these logical forms, when these are supported by concrete materials (Kuhn, 1977) or when they are supplied with additional information contradicting a biconditional interpretation and with training on producing uncertainty responses (Rumain et al., 1983). On the other hand, the idea that logical competence is acquired during adolescence implies that use of this competence should increase with age and experience. However, Klaczynski and Narasimham (1998) have

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observed a developmental decrease in logically correct responding to certain kinds of causal conditionals. In addition, several studies have shown that educated adults often produce incorrect biconditional responses to both affirmation of the consequent and denial of the antecedent (Cummins, 1995; Cummins et al., 1991; Markovits, 1985; Staudenmeyer & Bourne, 1977; Thompson, 1994). There are also many studies that have shown very large content-related effects in the way that both children and adolescents (Janveau-Brennan & Markovits, 1999; Klaczynski & Narasimham, 1998; Markovits, 2000; Markovits et al., 1998; Markovits & Vachon, 1990; Venet & Markovits, 2001) and adults (Cummins, 1995; Cummins et al., 1991; Quinn & Markovits, 1998; Thompson, 1994) reason with these forms. Thus, once again, there is a great deal of variability in the results of studies that have examined these more complex forms of reasoning. Although I have concentrated on the two invalid forms of conditional reasoning, it should be noted that similar degrees of variability are indeed observed across several kinds of reasoning tasks. For example, studies examining performance on the selection task have found very early abilities to produce logical responses on modified versions of this task (Cummins, 1996; Girotto & Light, 1993; Harris & Nùñez, 1996), others have observed a developmental increase in logical performance (Overton et al., 1987; Ward & Overton, 1990), while still others have observed very poor logical performance in educated adults (Evans, Newstead, & Byrne, 1993; Wason, 1968). This variability is found even for the metacognitive processes involved in the understanding of logical validity (Morris, 2000; Moshman & Franks, 1986) and the understanding of logical necessity (Miller, Custer, & Nassau, 2000; Shapiro & O'Brien, 1971).

Clearly, the inferential performance of both children and adults is strongly affected by both the content and the context of reasoning. There is strong evidence for both logical competence and logical incompetence in both children and adults on all of these forms of reasoning. Specifically, subsets of these results have been cited to support the following set of claims: (1) that very young children are logical in their responses, (2) that young children are not logical, but that reasoning ability develops in a way that leads to more logical performance, and (3) that both children and adults often do not reason logically.

These kinds of results pose clear problems for most of the developmental theories that have been described earlier. Almost all of these theories consider some level of logical competence to exist when reasoners can begin to respond normatively to some given set of inferential problems. While the notion of competence in developmental theories is complex, the simplest interpretation of this claim is that at some point in time, children or adolescents possess cognitive procedures that, when properly implemented, allow them to consistently respond logically to some set of inferences. A certain amount of variation in performance is not in itself a real problem,

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since the notion of competence does not imply total consistency. However, one would expect that at some point, the procedures that determine competence would be well established enough to allow a reasonable level of consistency. There is no real evidence that this is the case.

In fact, competence theories often attempt to minimize one or another of these claims. For example, neo-Piagetian theories such as Overton's generally suppose that young children's logical responses are the result of some, unspecified, nonformal procedure or heuristic. Early competence theories, such as that of Braine (1990), suppose that variation in performance is not related to logical competence, but only to peripheral, pragmatic factors. However, none of these attempts to reduce the scope of the extensive variations in performance are really satisfactory, particularly since these explanations rely, at least for the moment, on relatively unspecified mechanisms. And none of these theories can really explain how basic competence could be as fragile as the empirical results show them to be.

This basic problem is a direct consequence of the supposition that reasoning depends on the use of cognitive procedures that will invariably produce responses that are logical if correctly applied. If, in contrast, this assumption is not made, then what appears to be contradictory evidence can be reinterpreted very differently. Specifically, if it is supposed that the cognitive procedures that are used to make inferences in both children and adults have no necessary relation to logic, then the three conclusions stated previously can be rephrased in the following way: (1) the procedures used by young children to make inferences lead to responses that are, in some cases, the same as logical norms, (2) the procedures used by young children develop in such a way that the rate of responses that correspond to logical norms increases with age for at least some subset of inferential problems, and (3) the procedures used by both children and adults can lead to inconsistent responding on formally similar problems with variable content.

A DEVELOPMENTAL ADAPTATION OF JOHNSON-LAIRD'S MENTAL-MODEL THEORY

This suggests that one way to resolve the apparent contradiction in the empirical results would be to specify a procedural model that does not assume logic as a norm. Johnson-Laird's mental model theory (Johnson-Laird, 1983; Johnson-Laird & Byrne, 1991), which has been very successful in describing adult inferential performance, does just that. Briefly, this theory (see Johnson-Laird, Chapter 7 this book) supposes that reasoning requires the construction of a mental representation of the possible states of affairs described by the premises. Mental tokens are used to represent these states, and a process that requires both combining tokens and an active search for counterexamples is used for making inferences. Connectives are initially represented in a way that minimizes the cognitive load (the initial model)

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and if necessary are represented in a more complete way by a process that is referred to as fleshing out, which is the result of a semantic interpretation of the premises. The major constraint in reasoning concerns the necessity to retain several models in a limited working memory.

However, while mental-model theory has proven quite successful in describing the way that fairly complex reasoning is done by educated adults, the theory's developmental component is not clearly specified (Markovits & Barrouillet, in press). In fact, the adult version of this theory presupposes the existence of quite advanced cognitive processes that might not reasonably be available to children. The representations and procedures used in the theory correspond to what might well be considered to be quite high levels of symbolic processing; for example, the procedure hypothesized for generating complete sets of alternative models and the symbolic term used to indicate negation. Implicit to the theory is the hypothesis that all reasoners have access to the same basic procedures and symbolic representations irrespective of their level of expertise, and that imperfect reasoning is due to basically peripheral factors such as limitations in working-memory capacity.

Indeed, it could be claimed that mental-model theory appears to have logic as an implicit competence model, since the processes described by the theory will always lead to a logical conclusion if performed adequately. This is due, at least partly, to some basic characteristics of the theory; the relatively abstract form of the tokens that are used in models, the presence of an interpretative component (fleshing out) that often assigns a standard meaning to logical connectives in an unexplained manner, and the presence of an algorithm for searching for alternative models. For example, the theory assumes that tokens are basically abstract representations of premises. However, it is not clear that young children are capable of either generating or manipulating such kinds of symbols (see Sigel, 1999, for recent discussions of representational development). Similarly, the ability to search for alternative models requires a version of the kind of combinatorial ability that Piaget identified with formal thought. Research on the ability of children to systematically generate alternatives has shown that this can be very difficult even for adolescents (Roberge & Flexer, 1979). Markovits and Barrouillet (in press) have suggested a model in order to account for these developmental problems in a way that is both consistent with empirical data and that retains the adult version of mental-model theory as a developmental end point. In the following, I will present a brief version of some of the key points of this approach.

Firstly, mental tokens used by young children are assumed to refer to specific concrete elements (Barrouillet & Lecas, 1998; Markovits et al., 1996), that progress through a developmental sequence from referring to a specific element, to referring to classes of elements, and ultimately to more abstract elements (Venet & Markovits, 2001) via a redescriptive process

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such as that suggested by Karmilov-Smith (1992). Use of tokens that refer directly to specific elements (or classes of elements) in semantic memory provides for the concurrent activation of associated elements (Anderson, 1993). Markovits and Barrouillet (in press) have suggested that for most children and adolescents, models of premises are automatically fleshed out by a process of activation of information in long-term memory. The nature of the information that is activated depends on the specific connectives used, on the way that information has been encoded in memory, on task instructions, and on working-memory limitations.

The first major implication of these modifications is that when a child is in the process of attempting to make a conclusion on the basis of some major and minor premises, they have immediate access to whatever models have been generated by the activation process. This has implications for one of the key procedures postulated by mental-model theory; specifically, that putative conclusions for a given major and minor premise are evaluated by a two-pronged process, consisting of (1) generating additional models by an active search for alternatives and then (2) searching for models in which the minor premise is true. If there are any such models in which the putative conclusion is false and others in which this conclusion is true, then the reasoner will conclude that there is no certainty to this specific conclusion. Without the requirement to actively generate alternative models, only the second part of this process is necessary. This basically requires that children be able to detect uncertainty by searching for the presence of models with contradictory conclusions. This is well within children's competence. There have been many studies that have examined children's abilities to detect uncertainty. These have shown that young children can do so when the informational load of a given problem is relatively low and that there is a clear developmental increase in the ability of children to detect uncertainty in more complex situations (Fabricius, Sophian, & Wellman, 1987; Horobin & Acredolo, 1989; Sophian & Somerville, 1988; Wollman, Bat-Sheva, & Lawson, 1979).

A second important point is that fleshing out becomes a process that is constrained by two basic limitations. First, the nature of the information that is integrated into the fleshed-out models is determined by the nature and the structure of information that is stored in a reasoner's long-term memory. This translates the reasoner's experience with the connective and reflects the sum of the influences, both semantic and pragmatic, that are associated with use of the connective. Second, the specific content of fleshed-out models when making a specific inference is limited both by limited working memory and the efficiency of online information retrieval. And, as we shall see, these basic information processing constraints imply that inference making by younger reasoners will exhibit a form of the kind of local necessity which is implied by the Piagetian analysis of reasoning.

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At this point, it is useful to use a specific example to understand how this model might function. I will use the same two logical forms, modus ponens and affirmation of the consequent, that were examined previously. Both of these problems require conditional (if-then) reasoning. The first question that must be answered to see how this model might work concerns the semantic space that is determined by a reasoner's understanding of a conditional (if *p* then *q*) relation. Since even very young children have extensive experience with conditional statements of many forms (Scholnick, 1990), it can be assumed that this experience results in the generation of a semantic space that allows them to translate some of the basic structural characteristics of conditional relations. One key structural aspect, that reflects the way that conditionals are generally used, implies that the *p* term is generally understood as one possible object/event among many, while the *q* term produces a dichotomization of potential objects/events into *q* and *not-q*. For example, a reasoner who is given a premise such as "if it rains, then the street will be wet," will consider "it rains" as one event among many possible ones ("the sun shines," "it is cloudy," "a cleaner has passed," etc.). However, the consequent term allows for two basic possibilities, "the street is wet" or "the street is not wet." This basic decomposition of possible objects/events that are associated with the original premise is accompanied by the partitioning of the conceptual space of objects/events relevant to the conditional into three classes. The first concerns cases in which the objects or events concerned are complementary to those specified in the original conditional, that is, cases where objects or events that are different from *p* are related to *not-q* (we will refer to this as the complementary class). In the example we are using, this class would be composed of related events such as "if it is sunny, then the street will not be wet," or "if it is only cloudy, then the street will not be wet," etc.). The second class concerns possible objects/events that share the same relation to *q* as *p* does, that is, cases of *not-p* implies *q* (we will refer to this as the alternatives class). For example, "if the street cleaner passes, then the street will be wet" is one such example. Finally, the third class concerns what Cummins (1995) has called "disabling conditions," that is, conditions that allow the relationship between *p* and *q* to be violated (we will refer to this as the disabling class). For example, "if it rains, but the street is covered, then the street will not be wet" is one such example.

Now, what happens when a reasoner makes an inference that corresponds to the modus ponens form? Specifically, suppose that someone is given a problem of the form: Suppose that it is true that "if a rock is thrown at a window, the window will break. A rock is thrown at a window." The models that will be constructed by the reasoner will contain the model that is consistent with the major premise, which is the following:

Rock thrown at window – window broken

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Additional models will reflect the nature of information that is activated in memory, using the minor premise as an additional memory cue. In other words, the three classes of information that are cued by the child's basic semantic understanding of the if-then relation (complementary, alternatives, disabling) will be differentially activated by information contained in the minor premise. In this case, "a rock is thrown at a window" will most strongly cue for potential disabling conditions. If a disabling condition (e.g., the window is made of Plexiglas) is strongly activated, then a second model will be added to the initial one:

Rock thrown at Plexiglas window – window not broken

If only the initial model is generated, then the immediate conclusion that "*q* is true" will be drawn by the reasoner. If the two models are generated, then there is no conclusion to be drawn. Now, if it is reasonably assumed that young children have both less knowledge and more difficulty in accessing this knowledge (Kail, 1992), then a first consequence of this analysis is that young children would be expected to very often give what is in this case the logical response, that "*q* is true." Given that development is characterized by an increase in knowledge, and in particular an increase in knowledge of potential disabling conditions, and by an increase in facility in accessing this knowledge, then older reasoners would be expected to produce the two models more frequently, and thus to produce the nonlogical response that there is no certain conclusion more frequently than younger reasoners. In addition, specific content would be expected to have a strong effect on the kinds of inferences made to the modus ponens form, since it is relatively difficult to think of disabling conditions for a premise such as "all dogs have feet" than for a premise such as "if a stone is thrown at a window, the window will break." While a full explanation of the model would have to include the use of instructions to inhibit activation of potential disabling conditions (Markovits & Potvin, 2001; Vadeboncoeur & Markovits, 1999), this basic explanation clearly allows for both a high probability of very young children producing the logical response to many MP inferences, and for adults to have problems doing the same thing. It also clearly accounts for content variation, since the number of available disabling conditions is highly variable across contents (Cummins, 1995; Thompson, 1994).

A second example that can be used to illustrate the general developmental thrust of the model more completely is that of reasoning involving the affirmation of the consequent (AC) inference. Once again, I give a simplified version of the complete analysis. Making an AC inference requires reasoning with a "*p* implies *q*" major premise and a "*q* is true" minor premise. In this case, the minor premise will cue for both the complementary class and the alternatives class. If only information corresponding to the complementary class is highly activated, then the reasoner will produce the

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two following models:

Rock thrown at window	–	window broken
Rock not thrown at window	–	window not broken

In this case, the conclusion "that a rock was thrown at the window" follows from the minor premise "the window is broken." However, suppose that at least one example of the alternatives class is also highly activated, then the following three models will be produced:

Rock thrown at window	–	window broken
Rock not thrown at window	–	window not broken
Chair thrown at window	–	window broken

In this case, the reasoner will be able to conclude that there is no certain conclusion that follows from the minor premise "the window is broken." There are several interesting consequences of this explanation. Activation of relevant information is done online during the reasoning process, that is, when the reasoner is actively attempting to derive a conclusion. Thus, there is a concurrent cognitive load that corresponds to the effort required to keep all premises in working memory, to attention to instructions, etcetera, that limits the amount of resources available for searching through memory. This, in addition to the inherently probabilistic functioning of memory search implies that a reasoner will not consistently succeed in activating alternatives when reasoning with differing contents. In other words, if a reasoner is given a sequence of identical inferential problems, which vary only in the specific content of the premises, they will arrive at different conclusions, which has indeed been found to be true (Markovits, 1985). Another consequence of this is that content should affect the probability of activating a potential alternative in a way that is consistent with many observed results (Cummins, 1995; Markovits & Vachon, 1990; Thompson, 1994). More specifically, reasoning should reflect the way that information about alternatives is structured in long-term memory, something that has indeed been found to be true (Markovits et al., 1998; Quinn & Markovits, 1998). Finally, there are a number of interesting developmental consequences. When reasoning from concrete premises, younger children differ from older reasoners chiefly because they have a more limited working memory and relatively inefficient retrieval processes. When a young child constructs models of the premises, these are based on a relatively smaller search of their actual knowledge base than would be the case with older reasoners. A conclusion that is arrived at by a younger reasoner has a much higher probability of actually being an inaccurate reflection of the reasoner's own knowledge. Younger reasoners would thus be expected to produce conclusions that are only very locally valid, that is they are valid within only a small subset of available knowledge. For example, a young child has a fairly high probability of concluding that "if a window is broken,

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then a stone has been thrown at it" in the above example, despite their clear knowledge that things like chairs can also break windows (Markovits et al., 1998). This conclusion is considered valid at the moment of reasoning only because it is too difficult for the child to access all of his or her relevant knowledge during reasoning. At the moment that inference is made, the conclusion would be considered valid, although at another time, the same child might well activate a given alternative highly enough to generate a different conclusion to the same premises. Older reasoners would be able to access a greater portion of their knowledge, and their conclusions would thus be based on a much larger portion of their relevant knowledge. The chances of reaching a conclusion that was contradictory to stored knowledge would be less than for younger reasoners. Thus, in a very basic way, our mental-model analysis corresponds quite well to the Piagetian analysis of the relation between possibility and necessity which underlies the basic notion of formal thinking.

One other important developmental difference concerns the form of the models produced by reasoners. One of the key developmental processes that is postulated by Markovits and Barrouillet (in press) concerns the redescription of concrete elements into more abstract forms (Karmilov-Smith, 1992). Thus, younger children will use mostly specific concrete representations in their models of premises, but that older children will be able to represent classes (e.g., "things that can break window") and eventually, be able to use abstract representations as tokens (Markovits, Doyon, & Simoneau, in press; Venet & Markovits, 2001). Another possible difference might allow for a transition between an automatic activation process to a more combinatorial form of searching for alternative models that would, once again, provide a process that is consistent with the current formulation of the theory. These kinds of developmental changes would allow reasoning to more closely approximate what is considered to be logical, at least in circumstances where the reasoner adopted a more abstract form of reasoning.

I have only sketched an outline of what a fully developmental version of mental-model theory might look like. This description shows how this theory could account for observations (1) that young children can, in some cases, make inferences that correspond to logical norms, (2) that there is a developmental trend toward increased normative responding when more abstract modes of reasoning are employed, and an opposite trend in some cases when increased use of knowledge might lead to nonnormative responding, and (3) that there is a great deal of variability in reasoning at all developmental levels. While the details of this theory might well be put into doubt, it does seem to strongly suggest that the study of the development of reasoning might well profit from less attention to logical norms toward the more basic question of trying to understand the processes that underlie reasoning, logical or otherwise.

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